

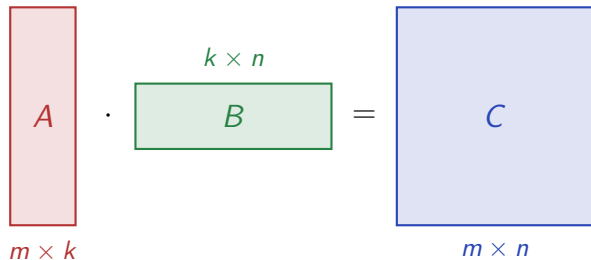
Cache-Optimal Tiling for Mixed-Precision Matrix Multiplication

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Matrix Multiply is Memory-Bound



for example:

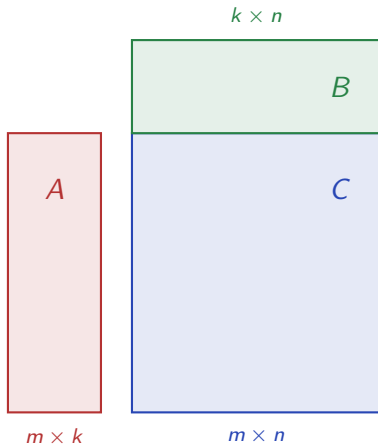
256×256 fp64: ≈ 0.5 MB

Typical L1 cache: 16–64 KB

$\Rightarrow 8 \times - 32 \times$ **larger than L1**

Fix: *tile* the loop — bring in one smaller C tile and keep it resident in L1.

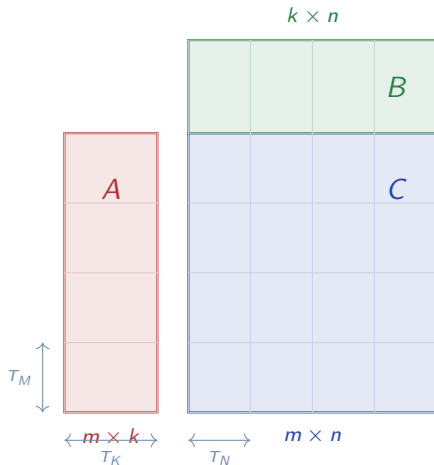
C-Stationary Tiling: Step by Step



$$C = A \cdot B$$

For large m, n, k — none fit in L1.

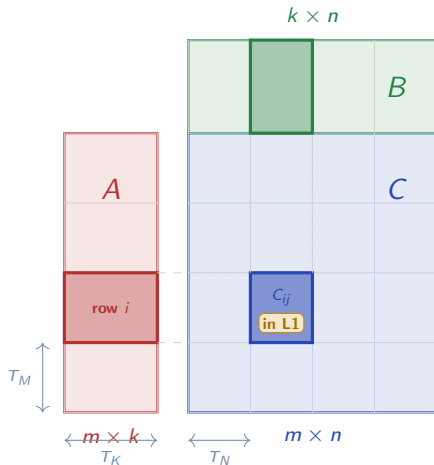
C-Stationary Tiling: Step by Step



Tile shape: $T_M \times T_N \times T_K$

Each matrix divided into a grid of tiles.

C-Stationary Tiling: Step by Step

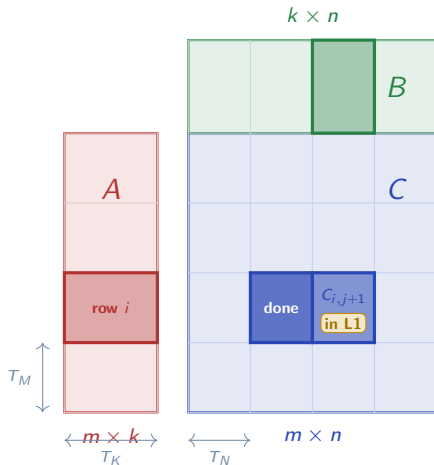


Fix C_{ij} in L1.

Stream A row-band and B col-band through it.

$$C_{ij} += A_{\text{row } i} \times B_{\text{col } j}$$

C-Stationary Tiling: Step by Step



C_{ij} done — move to B col $j+1$.

A row-band stays in L1.

Asymmetric Precision $\Rightarrow$ Asymmetric Tiles

$$\#A \text{ Bytes} = T_M \cdot k \cdot A_p$$

$$\#B \text{ Bytes} = T_N \cdot k \cdot B_p$$

$$\Rightarrow T = \frac{MN}{T_M T_N} (T_M \cdot k \cdot A_p + T_N \cdot k \cdot B_p)$$

$$T = MNK \left(\frac{A_p}{T_N} + \frac{B_p}{T_M} \right)$$

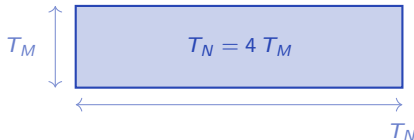
Find best ratio:

$$\Rightarrow \frac{T_N^*}{T_M^*} = \frac{A_p}{B_p} = \frac{1}{\rho}$$

$$\rho = 1 \text{ (fp32} \times \text{fp32)}$$


$$T_N$$
$$T_N$$

$$\rho = \frac{1}{4} \text{ (fp32} \times \text{fp8)}$$

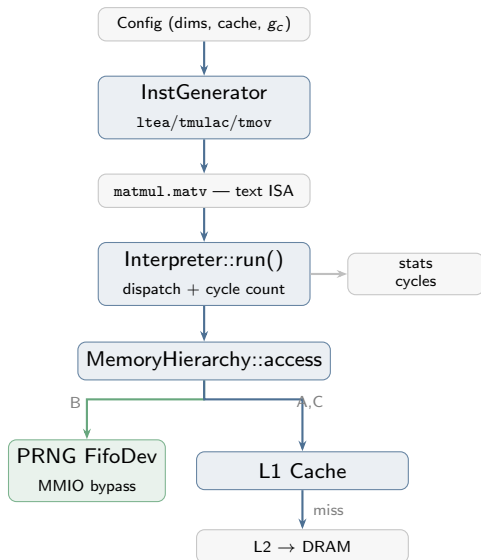


Same cache budget S — wider tile, better B reuse.

The Simulator We Built

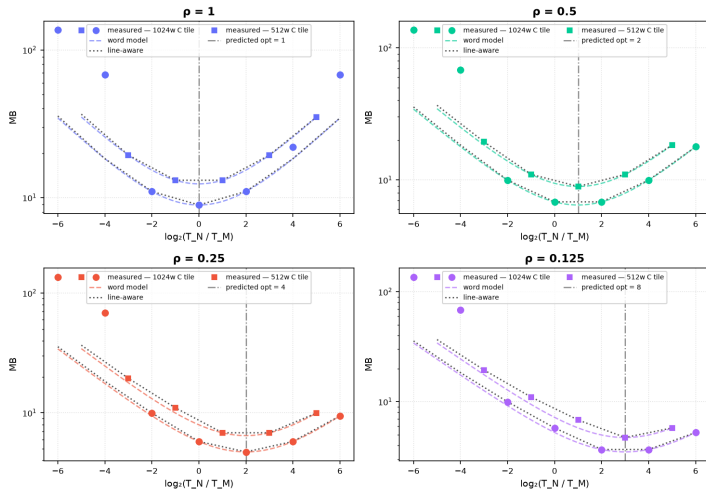
Three main components:

- **InstGenerator** — compiles tile ops to a text ISA (ltea/tmulac/tmov)
- **Interpreter** — dispatches instructions, counts cycles per access
- **MemoryHierarchy** — configurable L1/L2/DRAM with per-access tracing



Traffic Minimum at the Predicted Tile

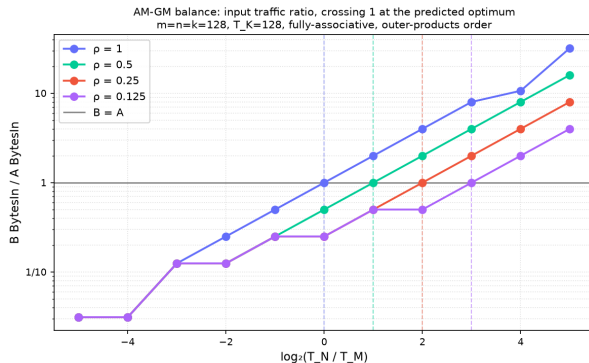
L1 reads (BytesIn) vs paper prediction — fully-associative cache
 $m=n=k=256$, $T_K=256$, outer-products order



Each curve shows L1 reads as a function of tile aspect ratio T_N/T_M .

Dashed line: paper's prediction
 $T_N/T_M = 1/\rho$.

B/A Balance: Exact Confirmation



Each curve is the ratio of B reads to A reads as T_N/T_M varies.

Traffic saved vs. square tile:

ρ	saving
1	0% (symmetric)
0.5	0%
0.25	18%
0.125	36%

The paper minimizes *bytes transferred*:

$$T_{\text{traffic}} = MNK \left(\frac{A_P}{T_N} + \frac{B_P}{T_M} \right)$$

But cycles $\neq$ bytes.

We need a *cycle-accurate* model.

Enter the PRNG FIFO

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B elements are generated *on-chip* by a hardware random number generator, streamed directly into the compute unit.

B never touches the cache.

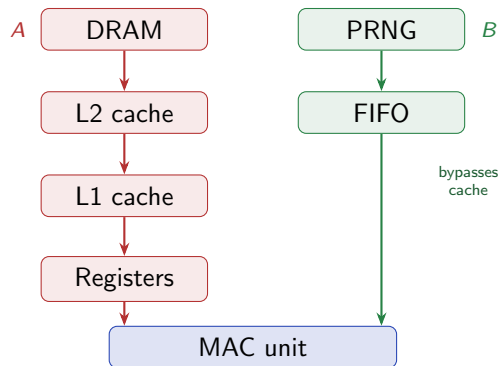
B has no memory address.

The B_P/T_M traffic term **vanishes**.

Instead, B introduces a new cost: on-chip *generation time*.

$$\text{generation cost} = g_c \text{ cycles / element}$$

The PRNG FIFO: B Without Memory



Random matmul: $C = A \cdot B$ where B is a pseudo-random matrix generated on-chip.

Use cases: random projections, privacy-preserving ML, approximate algorithms.

Two independent bottlenecks

A-bound: load A from memory

B-bound: wait for PRNG (g_c cy/elem)

The PRNG FIFO: B Without Memory

Runtime = ?

The PRNG FIFO: B Without Memory

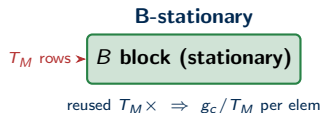
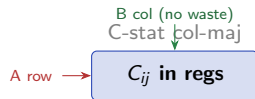
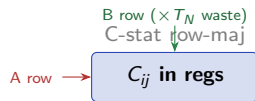
Runtime = whichever is slower

Which Loop Order Fits the FIFO?

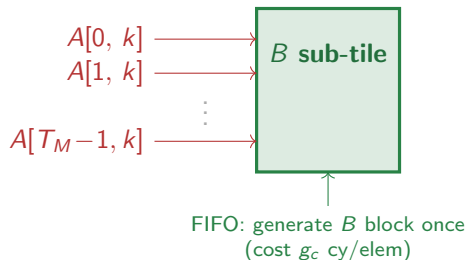
Problem: B elements emerge from the PRNG in a **fixed order** — you cannot skip or replay elements.

Three loop structures are possible:

Mode	In regs	B cost / elem
C-stat row-maj	C_{ij} tile	$g_c \times T_N$ (waste)
C-stat col-maj	C_{ij} tile	g_c
B-stationary	B_{ij} tile	g_c / T_M



B-Stationary: T_M -Fold Register Reuse



Loop order: B outer, A inner. Each B block is fetched *once*. $\rightarrow$ Each B element is used T_M times — generation cost amortized:

	$g_c=0$	$g_c=10$	$g_c=100$
B-stat	ref	ref	ref
C col-maj	$1.9 \times$ fstr	$1.25 \times$ fstr	$7.7 \times$ slwr
C row-maj	$1.6 \times$ slwr	$6.3 \times$ slwr	$61 \times$ slwr

Naive Cycle Model: Two Cost Terms

Same reuse argument as traffic, but in **cycles**:

Naively assuming the two costs **add**:

$$\frac{T}{MNK} = \frac{C_A}{T_N} + \frac{g_c}{T_M}$$

FIFO is Async: Two Costs in Parallel

The FIFO generates B **in the background** while the core loads A from L1.

The two costs **overlap** — runtime is determined by whichever finishes last:

$$\frac{T}{MNK} = \max \left\{ \frac{C_A}{T_N}, \frac{g_c}{T_M} \right\}$$

Two regimes:

$g_c/T_M < C_A/T_N$: A-load bound

$g_c/T_M > C_A/T_N$: B-gen bound

Replacing C_A/T_N with Measured $\alpha(T_M, T_N)$

The C_A/T_N term assumes:

C_A = fixed cost per A element from L1
 T_N -fold reuse divides it out cleanly

Problem: actual cycles/MAC depends on tile shape through cache residency, line utilization, and access patterns — not a simple constant divided by T_N .

Solution:

$$\frac{T}{MNK} = \max \left\{ \alpha(T_M, T_N), \frac{g_c}{T_M} \right\}$$

Isolating α : Set $g_c = 0$

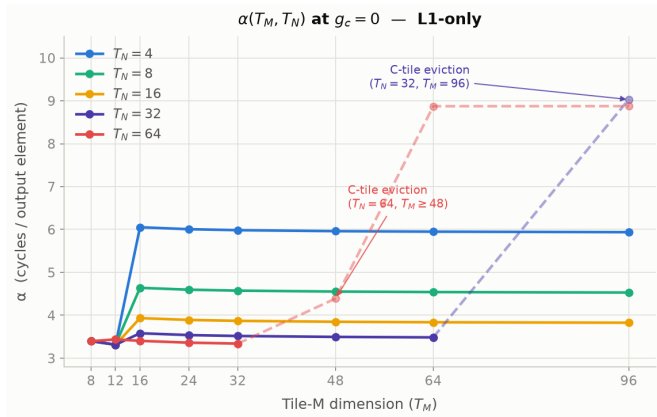
Key trick: set $g_c = 0$

The model collapses:

$$\frac{T}{MNK} = \max \left\{ \alpha(T_M, T_N), \frac{0}{T_M} \right\} = \alpha(T_M, T_N)$$

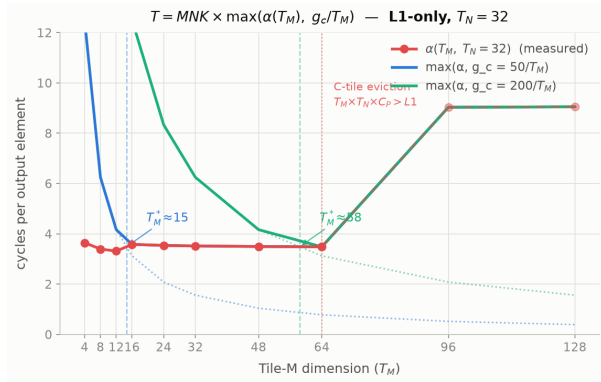
Calibration: Building the α Table

- Small T_N : α spikes when the A tile overflows L1
- Large T_N : α stays low until the C tile overflows L1
- α decreases with T_N (more column reuse)



Using the α Table: Predicting the Optimal Tile

The optimal tile is where the two costs are balanced: $\alpha(T_M^*, T_N^*) \approx g_c / T_M^*$.



Roofline Validation: Setup

Question:

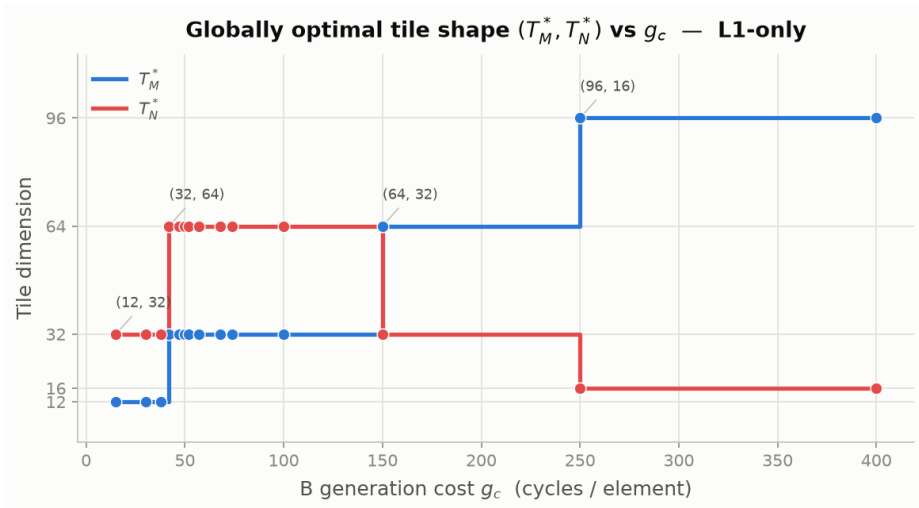
calibrate α once at $g_c = 0$; does the model predict the best tile correctly at any new g_c ?

Roofline Validation: Results

All 108 conditions	
(T_M^*, T_N^*) exact	108 / 108 (100%)

100% exact match across all hardware configurations and g_c values.

Impact of Tile Selection: Optimal vs. Square



Impact of Tile Selection: Optimal vs. Square

Question:

Does choosing the right (T_M, T_N) actually matter?

Comparing:

- Optimal asymmetric tile (T_M^*, T_N^*)
- Square tile ($T_M = T_N$, fixed)

Speedup from correct tile:

- $g_c = 100$: **20–65%** faster
- $g_c = 250$: **~85%** faster
- $g_c = 400$: **up to 90%** faster

Impact of Tile Selection: Optimal vs. Square

L1-only — empirically measured performance (E8 sweep)

